

Homework 5 Solutions

Math 117

March 31, 2026

1 Some exercises with linearity of expectation

- (a) The Massachusetts lottery has 1000 tickets. There is one ticket worth \$50000, nine worth \$2500, ninety worth \$250 each, and nine hundred worth \$25 each. What is the expected earnings from a randomly drawn ticket? What about five tickets?

Proof. Let X be the earnings from one ticket. Then,

$$\mathbb{E}[X] = \frac{1}{1000} \cdot 50000 + \frac{9}{1000} \cdot 2500 + \frac{90}{1000} \cdot 250 + \frac{900}{1000} \cdot 25 = \frac{235}{2} = \boxed{117.5}.$$

By linearity of expectation, the expected value from five tickets is $5 \cdot 117.5 = \boxed{587.5}$.

□

- (b) Suppose that in the lottery above, only 80% of tickets are sold (and the group of tickets sold is sampled uniformly at random). What is the expected amount to be awarded in total?

Proof. Let X be the total amount award. As per the hint, and by linearity of expectation,

$$\mathbb{E}[X] = 50000 \cdot \mathbb{E}[\mathbf{1}_{50000}] + \sum_{k=1}^9 2500 \cdot \mathbb{E}[\mathbf{1}_{2500,k}] + \sum_{j=1}^{90} 250 \cdot \mathbb{E}[\mathbf{1}_{250,j}] + \sum_{\ell=1}^{900} 25 \cdot \mathbb{E}[\mathbf{1}_{25,\ell}].$$

Each of these indicators are 0.8, as 80% of tickets are sold. So,

$$\mathbb{E}[X] = 0.8 \cdot (50000 + 2500 \cdot 9 + 250 \cdot 90 + 25 \cdot 900) = \boxed{94000}.$$

□

- (c) Six dice are rolled (not necessarily independently!). What is the expected value of the first roll? What is the expected sum of rolls?

Proof. The expected value of the first roll X_1 is

$$\mathbb{E}[X_1] = \frac{1+2+3+4+5+6}{6} = \frac{7}{2} = \boxed{3.5}.$$

Let $X_1, \dots, X_6$ be the six rolls. Then, by linearity of expectation,

$$\mathbb{E}[X_1 + \dots + X_6] = \mathbb{E}[X_1] + \dots + \mathbb{E}[X_6].$$

The expectations break up, regardless of their dependence or interdependence. Then, we can handle each independently. We know each one is 3.5, so the expected sum is $6 \cdot 3.5 = \boxed{21}$. □